X100/12/02

NATIONAL QUALIFICATIONS 2013

WEDNESDAY, 22 MAY
1.00 PM – 2.30 PM

MATHEMATICS
HIGHER
Paper 1
(Non-calculator)

Read carefully

Calculators may NOT be used in this paper.

Section A – Questions 1–20 (40 marks)

Instructions for completion of **Section A** are given on Page two.

For this section of the examination you must use an **HB pencil**.

Section B (30 marks)

1 Full credit will be given only where the solution contains appropriate working.

2 Answers obtained by readings from scale drawings will not receive any credit.

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

Read carefully

1 Check that the answer sheet provided is for **Mathematics Higher (Section A)**.

2 For this section of the examination you must use an **HB pencil** and, where necessary, an eraser.

3 Check that the answer sheet you have been given has **your name**, **date of birth**, **SCN** (Scottish Candidate Number) and **Centre Name** printed on it.

Do not change any of these details.

4 If any of this information is wrong, tell the Invigilator immediately.

5 If this information is correct, **print** your name and seat number in the boxes provided.

6 The answer to each question is **either** A, B, C or D. Decide what your answer is, then, using your pencil, put a horizontal line in the space provided (see sample question below).

7 There is **only one correct** answer to each question.

8 Rough working should **not** be done on your answer sheet.

9 At the end of the exam, put the **answer sheet for Section A inside the front cover of your answer book**.

Sample Question

A curve has equation $y = x^3 - 4x$.

What is the gradient at the point where $x = 2$?

A 8

B 1

C 0

D -4

The correct answer is **A—**8. The answer **A** has been clearly marked in **pencil** with a horizontal line (see below).



Changing an answer

If you decide to change your answer, carefully erase your first answer and, using your pencil, fill in the answer you want. The answer below has been changed to **D**.



Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

FORMULAE LIST

Circle:

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r.

Scalar Product: $\mathbf{a}.\mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

or $\mathbf{a}.\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$ where $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

Trigonometric formulae:

$$\begin{aligned} \sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\ \sin 2A &= 2\sin A \cos A \\ \cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A \end{aligned}$$

Table of standard derivatives:

$f(x)$	$f'(x)$
$\sin ax$	$a\cos ax$
$\cos ax$	$-a\sin ax$

Table of standard integrals:

$f(x)$	$\int f(x)dx$
$\sin ax$	$-\frac{1}{a}\cos ax + C$
$\cos ax$	$\frac{1}{a}\sin ax + C$

[Turn over

Looking for more resources? Visit scotpapers.com - Scotland's #1 Past Paper Archive

SECTION A

ALL questions should be attempted.

1. The functions f and g are defined by $f(x) = x^2 + 1$ and $g(x) = 3x - 4$, on the set of real numbers.

Find $g(f(x))$.

A $3x^2 - 1$

B $9x^2 - 15$

C $9x^2 + 17$

D $3x^3 - 4x^2 + 3x - 4$

2. The point P (5, 12) lies on the curve with equation $y = x^2 - 4x + 7$.

What is the gradient of the tangent to this curve at P?

A 2

B 6

C 12

D 13

3. Calculate the discriminant of the quadratic equation $2x^2 + 4x + 5 = 0$.

A –32

B –24

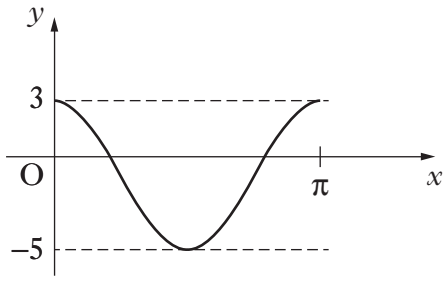
C 48

D 56

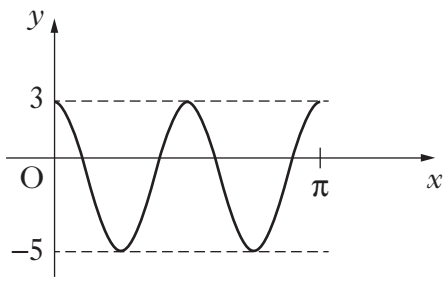
Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

4. Which of the following shows the graph of $y = 4\cos 2x - 1$, for $0 \leq x \leq \pi$?

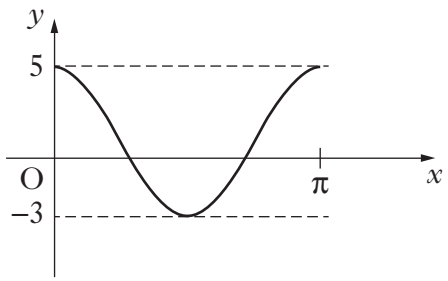
A



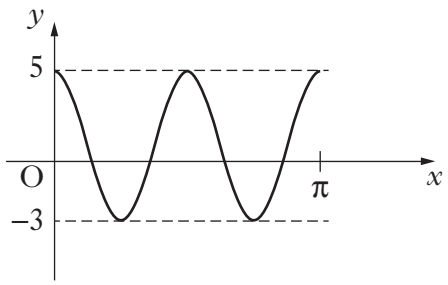
B



C



D



[Turn over

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

5. The line L passes through the point (–2, –1) and is parallel to the line with equation $5x + 3y - 6 = 0$.

What is the equation of L?

A $3x + 5y - 11 = 0$

B $3x + 5y + 11 = 0$

C $5x + 3y - 13 = 0$

D $5x + 3y + 13 = 0$

6. What is the remainder when $x^3 + 3x^2 - 5x - 6$ is divided by $(x - 2)$?

A 0

B 3

C 4

D 8

7. Find $\int x(3x + 2)\, dx$.

A $x^3 + c$

B $x^3 + x^2 + c$

C $\frac{1}{2}x^2\left(\frac{3}{2}x^2 + 2x\right) + c$

D $3x^2 + 2x + c$

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

8. A sequence is defined by the recurrence relation $u_{n+1} = 0{\cdot}1u_n + 8$, with $u_1 = 11$.

Here are two statements about this sequence:

(1) $u_0 = 9{\cdot}1$;

(2) The sequence has a limit as $n \longrightarrow \infty$.

Which of the following is true?

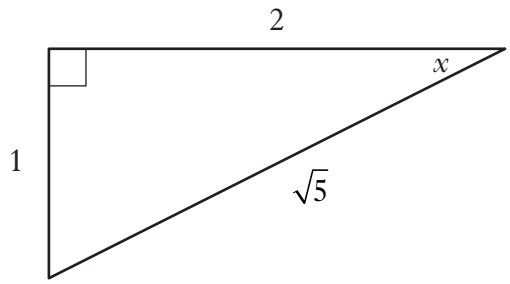
A Neither statement is correct.

B Only statement (1) is correct.

C Only statement (2) is correct.

D Both statements are correct.

9. The diagram shows a right-angled triangle with sides and angles as marked.



Find the value of $\sin 2x$.

A $\frac{4}{5}$

B $\frac{2}{5}$

C $\frac{2}{\sqrt{5}}$

D $\frac{1}{\sqrt{5}}$

10. If $0 < a < 90$, which of the following is equivalent to $\cos(270 - a)°$?

A $\cos a°$

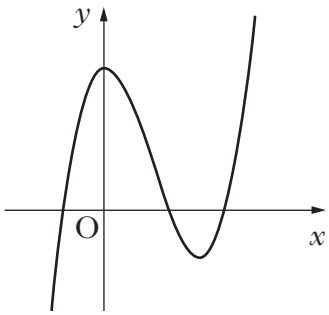
B $\sin a°$

C $-\cos a°$

D $-\sin a°$

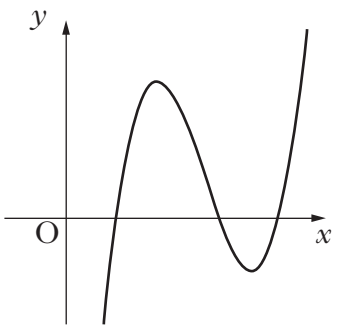
Looking for more resources? Visit scotpapers.com - Scotland's #1 Past Paper Archive

11. The diagram shows a cubic curve with equation $y = f(x)$.

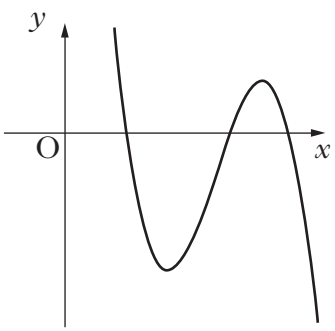


Which of the following diagrams could show the curve with equation $y = -f(x - k)$, $k > 0$?

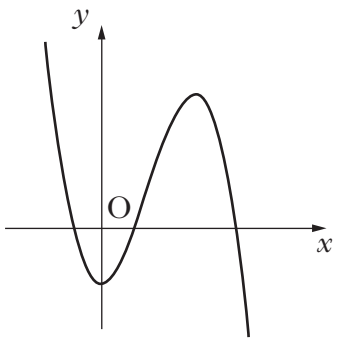
A



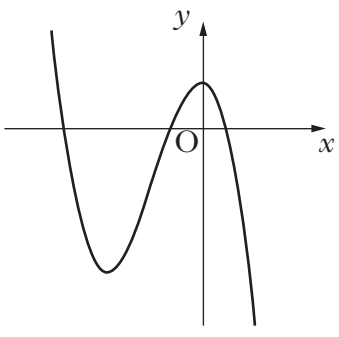
B



C



D



Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

12. If $\mathbf{f} = 3\mathbf{i} + 2\mathbf{k}$ and $\mathbf{g} = 2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$, find $|\mathbf{f} + \mathbf{g}|$.

A $\sqrt{14}$ units

B $\sqrt{42}$ units

C $\sqrt{66}$ units

D $\sqrt{70}$ units

13. A function f is defined on a suitable domain by $f(x) = \dfrac{x+2}{x^2 - 7x + 12}$.

What value(s) of x cannot be in this domain?

A 3 and 4

B -3 and -4

C -2

D 0

14. Given that $|\mathbf{a}| = 3$, $|\mathbf{b}| = 2$ and $\mathbf{a}.\mathbf{b} = 5$, what is the value of $\mathbf{a}.(\mathbf{a} + \mathbf{b})$?

A 11

B 14

C 15

D 21

15. Solve $\tan\left(\frac{x}{2}\right) = -1$ for $0 \leq x < 2\pi$.

A $\dfrac{\pi}{2}$

B $\dfrac{7\pi}{8}$

C $\dfrac{3\pi}{2}$

D $\dfrac{15\pi}{8}$

[Turn over

Looking for more resources? Visit scotpapers.com - Scotland's #1 Past Paper Archive

16. Find $\int (1-6x)^{-\frac{1}{2}} dx$ where $x < \frac{1}{6}$.

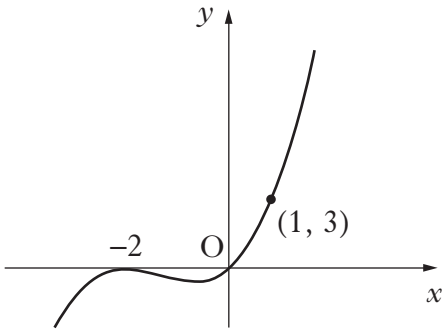
A $\frac{1}{9}(1-6x)^{-\frac{3}{2}} + c$

B $3(1-6x)^{-\frac{3}{2}} + c$

C $-\frac{1}{3}(1-6x)^{\frac{1}{2}} + c$

D $-3(1-6x)^{\frac{1}{2}} + c$

17. The diagram shows a curve with equation of the form $y = kx(x + a)^2$, which passes through the points (–2, 0), (0, 0) and (1, 3).



What are the values of a and k?

	a	k
A	–2	$\frac{1}{3}$
B	–2	3
C	2	$\frac{1}{3}$
D	2	3

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

18. Given that $y = \sin(x^2 - 3)$, find $\frac{dy}{dx}$.

A $\sin 2x$

B $\cos 2x$

C $2x \sin(x^2 - 3)$

D $2x \cos(x^2 - 3)$

19. Solve $1 - 2x - 3x^2 > 0$, where x is a real number.

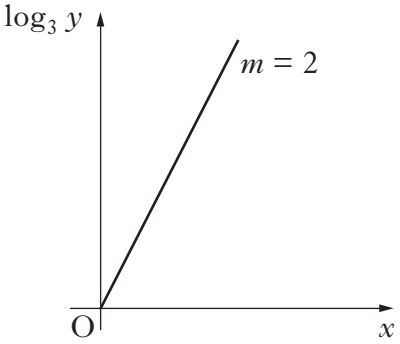
A $x < -1$ or $x > \frac{1}{3}$

B $-1 < x < \frac{1}{3}$

C $x < -\frac{1}{3}$ or $x > 1$

D $-\frac{1}{3} < x < 1$

20. The graph of $\log_3 y$ plotted against x is a line through the origin with gradient 2, as shown.



Express y in terms of x.

A $y = 2x$

B $y = 9x$

C $y = 6^x$

D $y = 9^x$

[*END OF SECTION A*]

[Turn over for SECTION B on *Page twelve*]

Looking for more resources? Visit scotpapers.com - Scotland's #1 Past Paper Archive

SECTION B

Marks

ALL questions should be attempted.

21. Express $2x^2 + 12x + 1$ in the form $a(x + b)^2 + c$. **3**

22. A circle C_1 has equation $x^2 + y^2 + 2x + 4y - 27 = 0$.

(*a*) Write down the centre and calculate the radius of C_1. **2**

(*b*) The point P(3, 2) lies on the circle C_1.
Find the equation of the tangent at P. **3**

(*c*) A second circle C_2 has centre (10, –1). The radius of C_2 is half of the radius of C_1.
Show that the equation of C_2 is $x^2 + y^2 - 20x + 2y + 93 = 0$. **3**

(*d*) Show that the tangent found in part (*b*) is also a tangent to circle C_2. **4**

23. (*a*) The expression $\sqrt{3}\sin x° - \cos x°$ can be written in the form $k \sin(x - a)°$, where $k > 0$ and $0 \leq a < 360$.
Calculate the values of k and a. **4**

(*b*) Determine the maximum value of $4 + 5\cos x° - 5\sqrt{3}\sin x°$, where $0 \leq x < 360$. **2**

24. (*a*) (i) Show that the points A(–7, –8, 1), T(3, 2, 5) and B(18, 17, 11) are collinear.

(ii) Find the ratio in which T divides AB. **4**

(*b*) The point C lies on the x-axis.
If TB and TC are perpendicular, find the coordinates of C. **5**

[*END OF SECTION B*]

[*END OF QUESTION PAPER*]

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive